

Pengantar Embedded System & Mikrokontroler

IFE307 — Embedded System

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Apa itu Embedded System?

Sistem komputer yang dirancang untuk satu fungsi spesifik, tertanam di dalam perangkat yang lebih besar.

- Tidak seperti PC yang *general purpose*
- Bekerja dalam **constraint** ketat: daya, ukuran, real-time
- Mayoritas "komputer" di dunia adalah embedded system

Contoh cepat: ketika kamu menekan tombol lift di Gedung J , ada mikrokontroler yang memprosesnya – bukan laptop.

Embedded System Ada di Mana-Mana

Sektor	Contoh Perangkat
Rumah tangga	Mesin cuci, AC, microwave, kulkas
Otomotif	ABS, ECU mesin, airbag controller
Medis	Pacemaker, infusion pump, glucometer
Industri	PLC, robot arm, conveyor control
Konsumer	Kamera digital, smartwatch, TWS earbuds

Estimasi: lebih dari 98% dari seluruh prosesor yang diproduksi adalah untuk embedded system.

Aplikasi Smart System

Embedded System sebagai Tulang Punggung Smart System

Embedded system adalah fondasi dari seluruh ekosistem smart technology.

Smart Home:

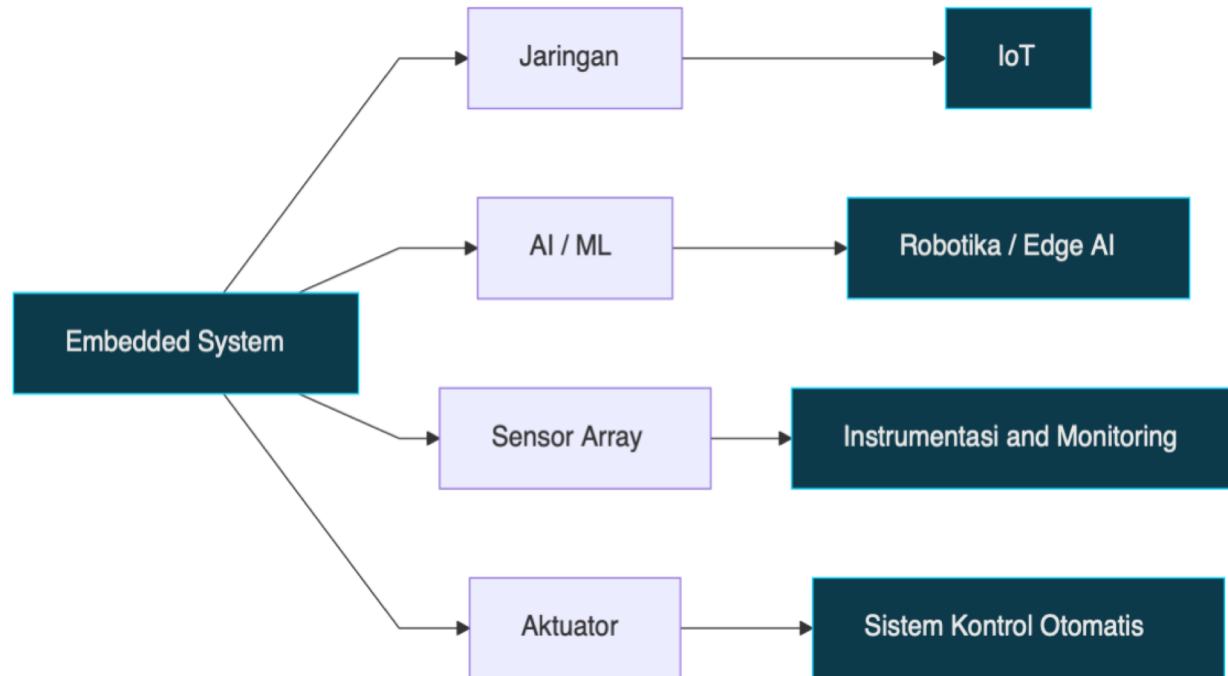
- Pengatur suhu adaptif (thermostat pintar)
- Lampu yang menyesuaikan intensitas otomatis
- Kunci pintu berbasis fingerprint / RFID

Smart City:

- Smart parking – sensor mendeteksi ketersediaan slot
- Pemantau kualitas udara real-time
- Intelligent Transportation System (ITS)

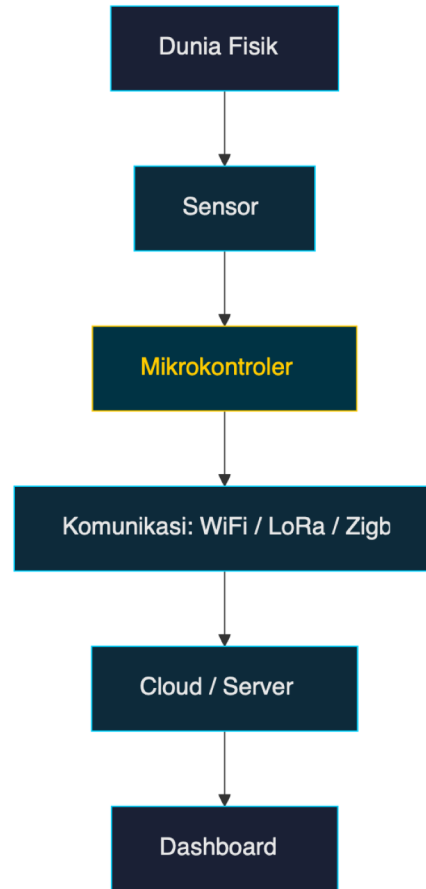
Ribuan sensor bekerja secara **real-time**, dikumpulkan, diproses, dan divisualisasikan

Posisi dalam Ekosistem Teknologi



- **IoT** = Embedded System yang terhubung ke internet dan saling berkomunikasi
 - **Edge AI** = model AI yang berjalan langsung di mikrokontroler, tanpa cloud
 - **Robotika** = embedded yang mengontrol gerak, persepsi, dan keputusan
- IoT punya kuliah tersendiri – di sini kita bangun **fondasinya**.

Anatomi Node IoT



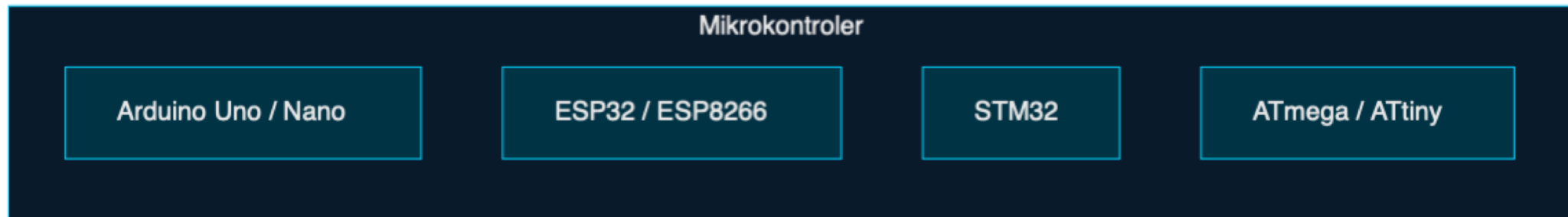
- Raspberry Pi, ESP32, Arduino → **node IoT**.

Mikrokontroler vs Mikroprosesor

	Mikrokontroler	Mikroprosesor
Contoh	ATmega328P, STM32	Intel Core i7, AMD Ryzen
Memori	Built-in (Flash, SRAM, EEPROM)	Eksternal (RAM, SSD)
Performa	Ratusan MHz	GHz
Daya	Mili-watt	Watt hingga puluhan watt
OS	Bare metal / RTOS	Linux, Windows
Harga	Di bawah \$5	Ratusan ribu rupiah
Kegunaan	Kontrol spesifik	Komputasi umum

Mikrokontroler bukan "lebih buruk" – ia **tepat guna** untuk tugasnya.

Platform Populer dalam Embedded System



- SBC: OS penuh, lebih powerful, butuh lebih banyak daya.
- Mikrokontroler: bare metal, hemat daya, harga murah

Arduino



ARDUINO YUN



ARDUINO ROBOT



ARDUINO MEGA ADK



ARDUINO LEONARDO



ARDUINO DUE



ARDUINO ETHERNET



ARDUINO ESPLORA



ARDUINO MINI



ARDUINO FIO



ARDUINO NANO



ARDUINO GSM SHIELD



ARDUINO ETHERNET SHIELD



Arduino Uno R3



Arduino Uno R3 SMD



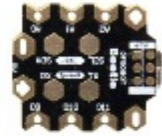
Arduino Pro



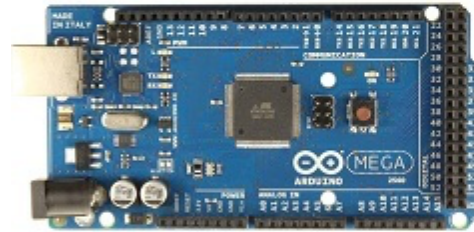
Arduino Zero



Arduino Ethernet



Arduino Beetle



Arduino Mega 2560



Arduino Nano



Arduino Fio



Arduino ADK



Arduino Leonardo

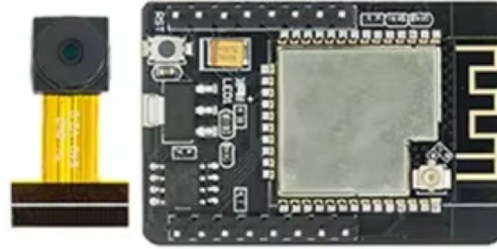
ESP



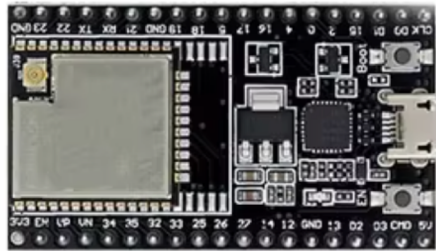
ESP-32 30PIN



ESP-32 38PIN



ESP32-CAM



ESP32-WROOM-32U



ESP32-WROOM-32D

Raspberry Pi

Raspberry Pi 1 Model B+



Raspberry Pi 1 Model A+



Raspberry Pi 2 Model B



Raspberry Pi 3 Model B



Raspberry Pi 3 Model A+



Raspberry Pi 3 Model B+



Raspberry Pi 4 Model B



Raspberry Pi Zero 1.3



Raspberry Pi Zero W



Raspberry Pi Zero 2 W



Raspberry Pi 5

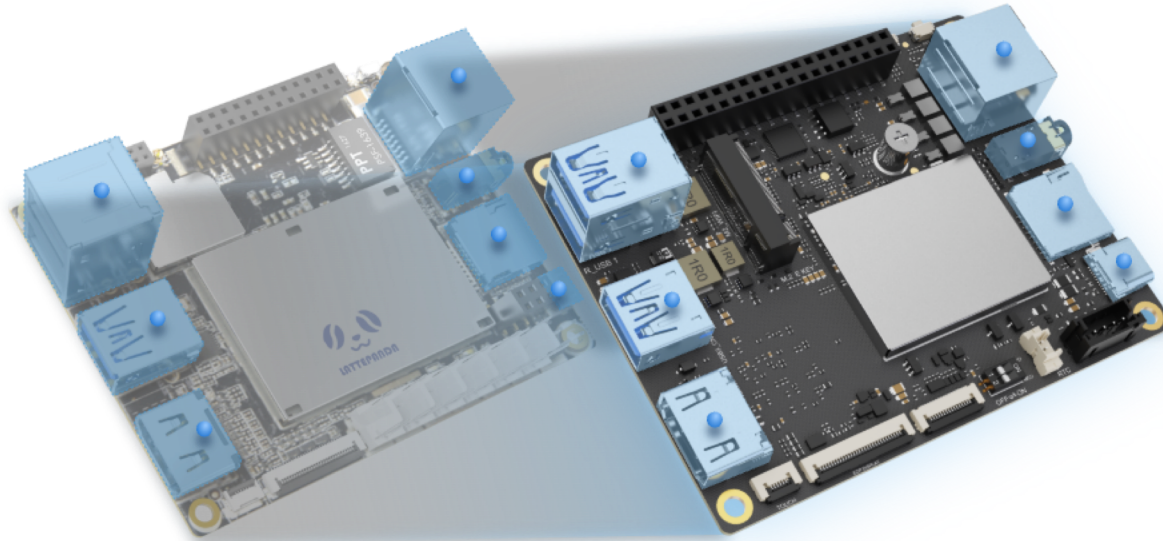


RaspberryPi



NextPCB
www.nextpcb.com

LattePanda



 Intel Atom x5-Z8350 up to 4 Core 4 Thread 1.92 GHz CPU	LattePanda V1 [2G RAM 32G eMMC] www.lattepanda.com PC01D2A03MLPDFR0416	88mm x 70mm Credit Card Size
2 / 4GB Memory	Fanless 2W TDP Low Power Consumption	
32 / 64GB Storage	 Windows	 Ubuntu
 USB 3.0	 GPIO	 Wi-Fi 4
 Bluetooth 4	 HDMI	 MIPI Touch Display
 Ethernet		

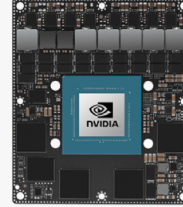
 Intel N150	4 Core 4 Thread Up to 3.6 GHz Processor	88mm x 70mm Palm-Sized
8 / 16 GB Memory	 LattePanda IOTA	6 ~ 15W Configurable TDP
64 / 128 GB Storage	 Windows	 Ubuntu
10Gbps 3 x USB 3.2	RP2040 Co-processor	PCIe 3.0 x1 FPC
	 eDP Touch Display	 M.2 E Key

Jetson



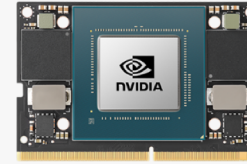
Jetson Thor series

Up to 2070 TOPS
40-130 W
100 mm x 87 mm



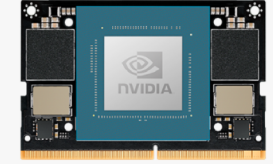
Jetson AGX Orin series

Up to 275 TOPS
15-60 W
100 mm x 87 mm



Jetson Orin NX series

Up to 157 TOPS
10-25 W
70 mm x 45 mm



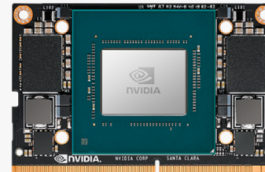
Jetson Orin Nano™ series

Up to 67 TOPS
7-15 W
70 mm x 45 mm



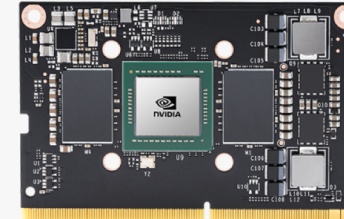
Jetson AGX Xavier™ series

Up to 32 TOPS
10-30 W | 20-40 W
100 mm x 87 mm



Jetson Xavier NX series

21 TOPS
10-20 W
70 mm x 45 mm



Jetson TX2 series

1.3 TFLOPS
7.5-15 W | 10-20 W
70 mm x 45 mm



Jetson Nano™

0.5 TFLOPS
5-10W
70mm x 45mm

Ekosistem Hardware Embedded System

Mikrokontroler (Low-power, task-specific):

- **Arduino Uno/Nano** – entry level, ideal untuk belajar
- **ESP32** – Arduino + WiFi + Bluetooth built-in
- **STM32** – lebih kuat, dipakai di industry

Single Board Computer / SBC (lebih powerful):

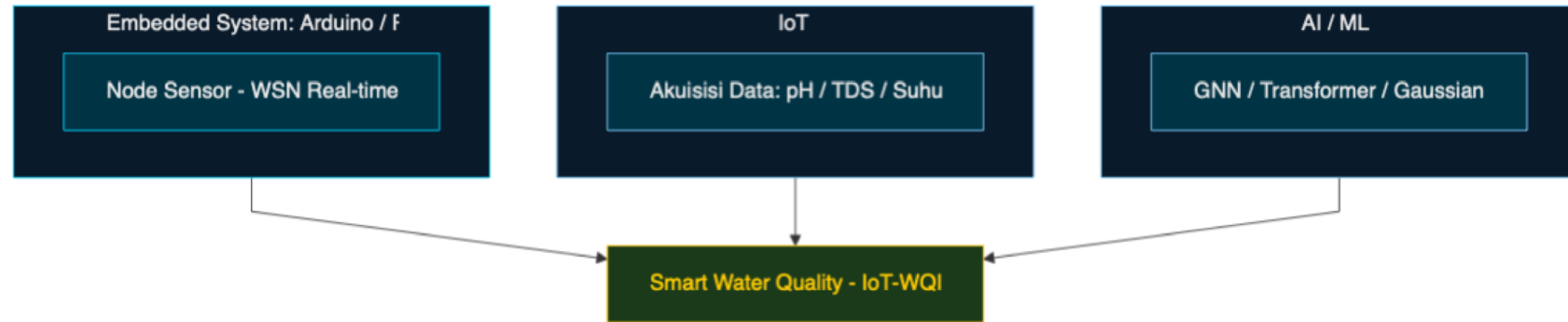
- **Raspberry Pi** – Linux-based, cocok untuk AI inference, server lokal
- **LattePanda** – SBC berbasis Windows, Intel CPU, cocok prototipe industri
- **NVIDIA Jetson Nano** – GPU onboard, edge AI & computer vision

Perbedaan utama: SBC menjalankan **OS penuh**, mikrokontroler menjalankan **bare metal atau RTOS**.

Kapan Pakai Apa?

Kebutuhan	Pilihan
Baca sensor sederhana, kontrol relay	Arduino Uno/Nano
Koneksi WiFi + sensor	ESP32
Streaming video + object detection	Raspberry Pi
Prototipe industri dengan Windows	LattePanda
Real-time edge AI	Jetson Nano

Ekosistem Hardware Embedded System



Fase	Periode	Fokus
Fondasi	2013	Kendali Smart Home dengan IoT
	2014-2016	Embedded sensor, WSN, database real-time Carbon Monoxide
IoT Applied	2018-2020	Water quality monitoring (DO, pH, TDS, turbidity)
Pemodelan	2020-2025	IoT-WQI, Indeks Kualitas Air
AI/ML	2025-2026	GNN, Transformer, Edge AI, XAI, TSVG HGBR, STL HGBR, GIFT, ViT-GNN RFFA,

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